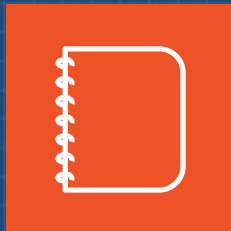


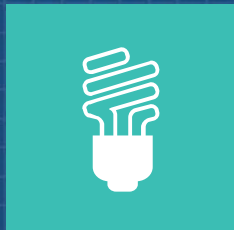
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2012



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DEV212: Implementing Scrum Using Team Foundation Server 11 2012



Richard Hundhausen
President, Accentient

Richard Hundhausen

- ④ From Boise, Idaho (father of five)
- ④ President of Accentient
- ④ Professional Scrum Trainer
- ④ Microsoft RD & MVP (Visual Studio ALM)
- ④ Author of Professional Scrum Developer course and Professional Scrum Development Using Visual Studio 2012
- ④ richard@accentient.com
- ④  @rhundhausen



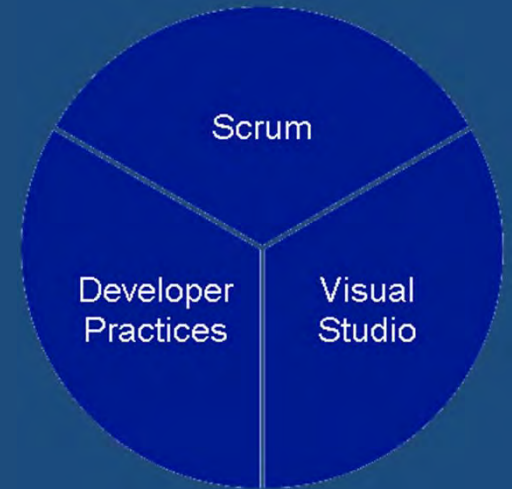
Accentient

- ④ Is a leader in Visual Studio ALM knowledge
- ④ Has helped hundreds of clients understand, implement and leverage Team Foundation Server
- ④ Has a close working relationship with Microsoft
- ④ Has a cross-functional team consisting of Microsoft MVPs, Microsoft RDs, Professional Scrum Trainers, and Professional Scrum Developers

www.accentient.com

Scrum.org

- ④ Scrum.org is the home of Scrum
 - ④ Knowledge, training, and assessments
- ④ Mission:
 - ④ *To Improve the Profession of Software Development*



#proscrumdev | @scrumdotorg

Session Backlog

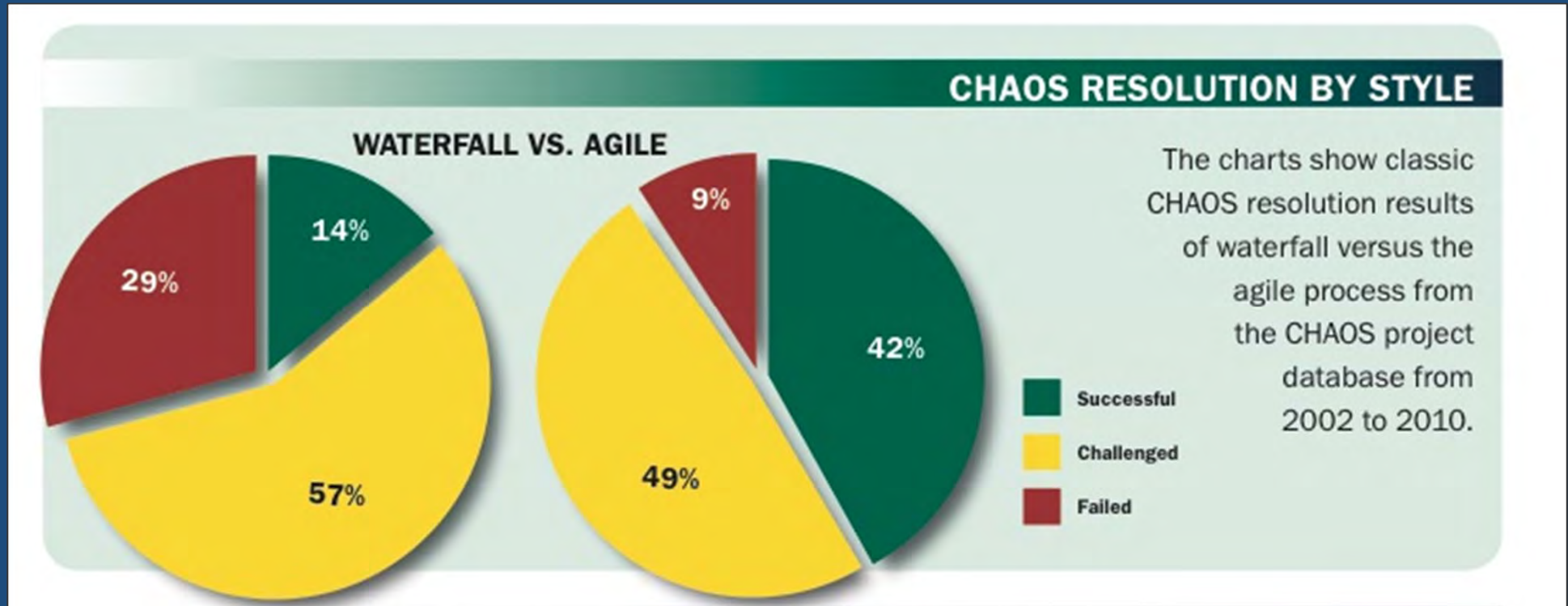
- ④ Why Scrum?
- ④ Scrumdamentals
- ④ Visual Studio Scrum 2.0
- ④ The Product Backlog
- ④ The Sprint
- ④ Getting to "Done"

Why Scrum?



(Because it works)

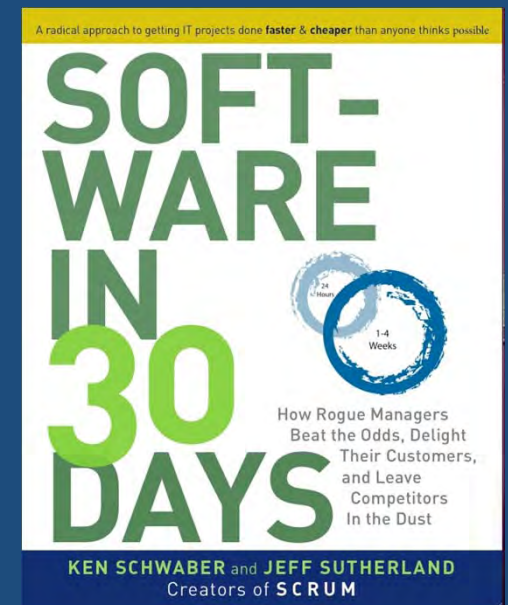
The CHAOS Manifesto (2011)



The CHAOS Manifesto, Copyright 2011

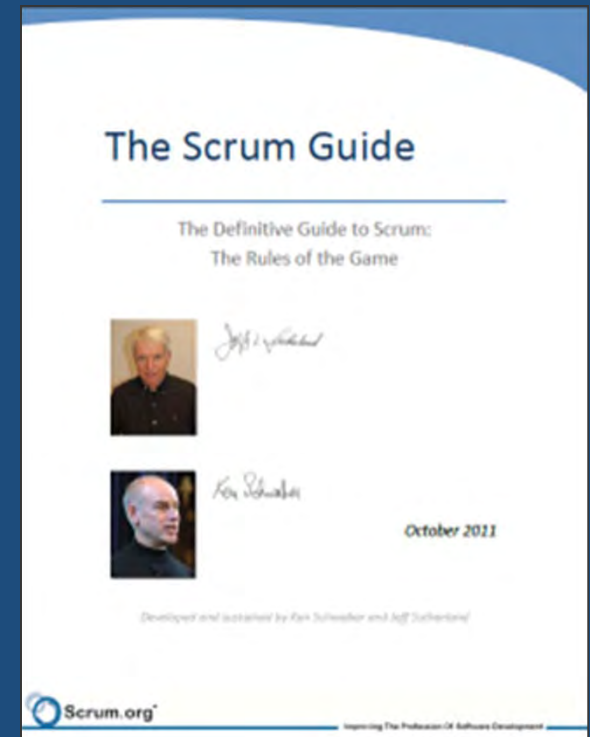
FBI Automated Case Support (ACS) program

- ④ A "CRM" for criminals built using 70s era tools
- ④ Was to be replaced by *Virtual Case File*
 - ④ 4 years and \$170 million wasted
- ④ Was to be replaced by Sentinel
 - ④ 4 years and \$ 405 million wasted
- ④ Project reset using Scrum
 - ④ Reduced staff (400 to 40)
 - ④ 1 year and \$ 30 million: Code complete



Scrum is ...

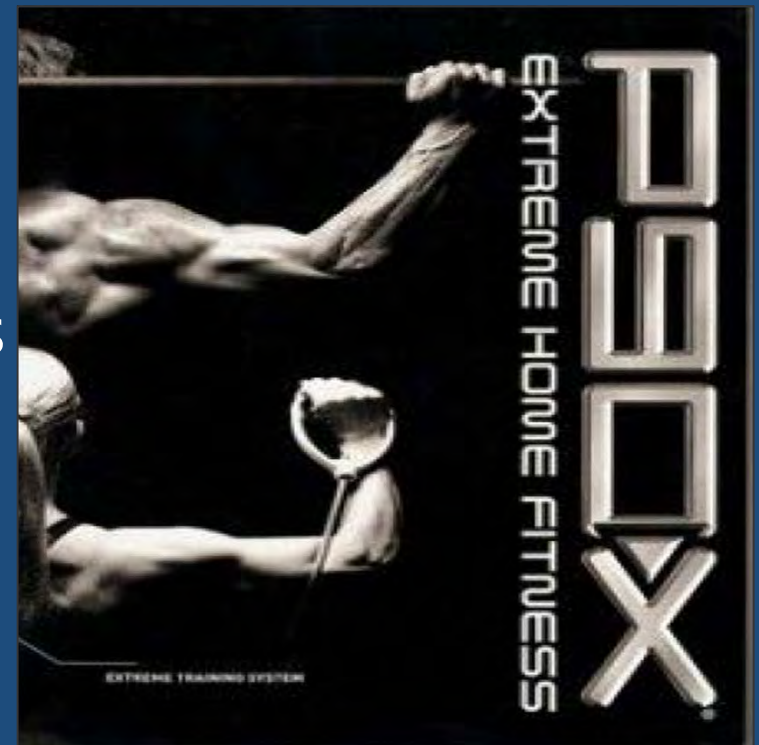
- ① A framework for Agile software development
- ① A set of rules defined in the Scrum Guide
- ① Easy to learn, difficult to master



<http://www.scrum.org>

Scrum is to Agile ...

- ④ As P90X® is to “getting healthy”
- ④ Both have core principles:
 - Eat better and exercise
- ④ P90X® is a framework:
 - Daily menus and specific exercises
- ④ In both cases, you are still the one that has to do it every day!





Scrumdamentals



(This is all in the Scrum Guide)

Roles, Events, Artifacts

Roles

Product Owner

Scrum Master

Development
Team

Events

The Sprint

Sprint Planning
Meeting

Daily Scrum

Sprint Review

Sprint Retrospective

Note: All Events are timeboxed

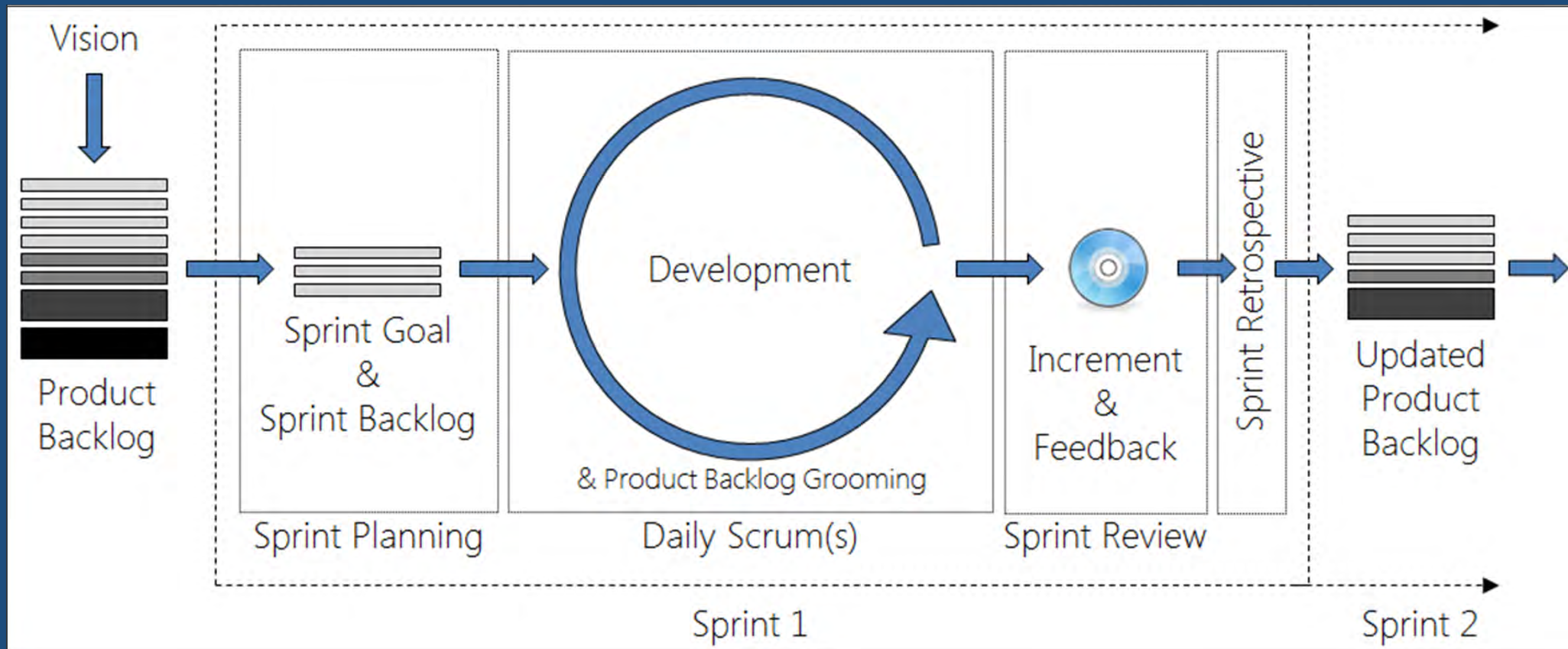
Artifacts

Product Backlog

Sprint Backlog

The Increment
(of working software)

The Scrum Events and Artifacts



Role: Product Owner

- ④ The Product Owner is responsible for maximizing the value of the product and the work of the Development Team

Role: Scrum Master

- ④ The Scrum Master is responsible for ensuring Scrum is understood and enacted

Role: Development Team

- ④ The Development Team consists of professionals who do the work of delivering a potentially releasable Increment of “Done” product at the end of each Sprint

Event: **The Sprint**

- ④ The heart of Scrum is a Sprint, a time-box of one month or less during which a “Done”, useable, and potentially releasable product Increment is created

Event: Sprint Planning Meeting

- ④ The work to be performed in the Sprint is planned at the Sprint Planning Meeting and the plan is created by the collaborative work of the entire Scrum Team

Event: Daily Scrum

- ④ The Daily Scrum is a 15-minute time-boxed event for the Development Team to synchronize activities and create a plan for the next 24 hours

Event: Sprint Review

- ④ A Sprint Review is held at the end of the Sprint to inspect the Increment and adapt the Product Backlog if needed

Event: Sprint Retrospective

- ④ The Sprint Retrospective is an opportunity for the Scrum Team to inspect itself and create a plan for improvements to be enacted during the next Sprint

Artifact: Product Backlog

- ④ The Product Backlog is an ordered list of everything that might be needed in the product and is the single source of requirements for any changes to be made to the product

Artifact: **Sprint Backlog**

- ④ The Sprint Backlog is the set of Product Backlog items selected for the Sprint plus a plan for delivering the product Increment and realizing the Sprint Goal

Artifact: **The Increment**

- ④ The Increment is the sum of all the Product Backlog items completed (according to the definition of “Done”) during a Sprint and all previous Sprints



Visual Studio Scrum



(100,000+ downloads of v1.0)

Visual Studio Scrum 2.0

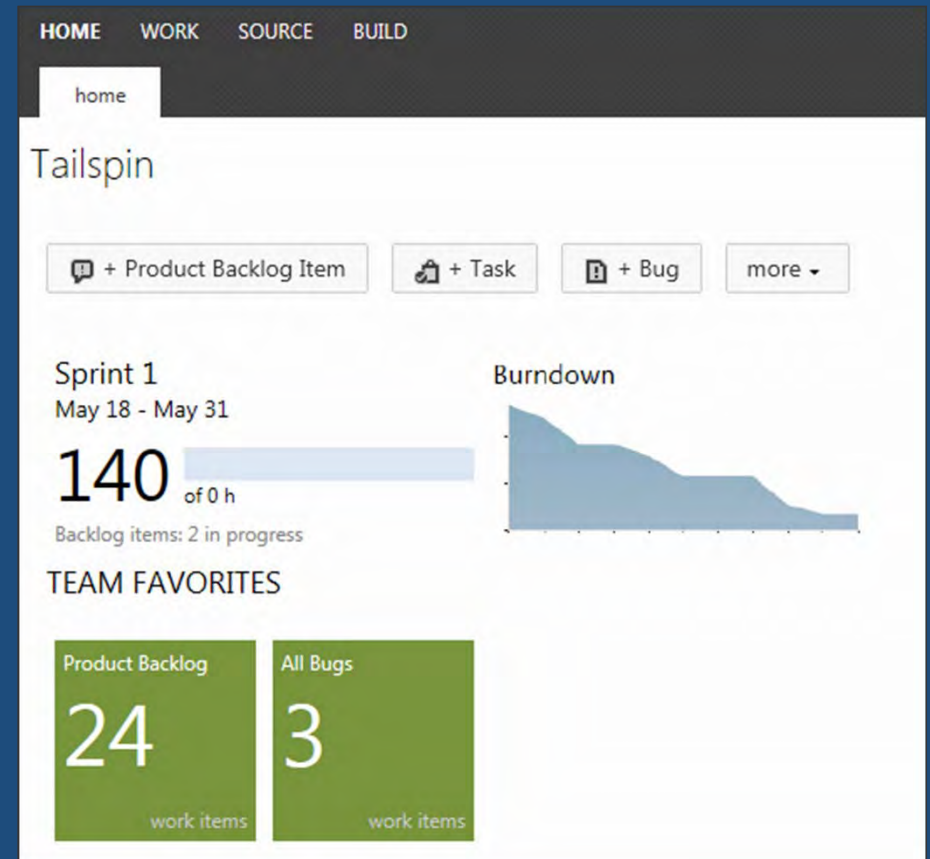
- ④ Visual Studio Scrum 1.0 was introduced shortly after Visual Studio 2010 launched
- ④ Visual Studio Scrum 2.0 will be the default process template
 - ④ In both hosted Team Foundation Service and on-premises TFS
 - ④ This demonstrates how popular Scrum is and also Microsoft's commitment to supporting Scrum
- ④ The template maps directly to the Scrum concepts
 - ④ Sprint vs. Iteration, PBI vs. User Story, Impediment vs. Issue, etc.

Visual Studio Scrum 2.0

Work Item Types	Queries	Reports
Product Backlog Item	Product Backlog	Backlog Overview *
Task	Feedback Requests *	Release Burndown
Code Review Request *	Current Sprint	Sprint Burndown
Code Review Response *	Blocked Tasks	Velocity
Feedback Request *	Open Impediments	Build Success Over Time
Feedback Response *	Sprint Backlog	Build Summary
Impediment	Test Cases	Test Case Readiness
Test Case	Unfinished Work	Test Plan Progress
Shared Steps	Work in Progress	
Sprint *		

Team Web Access

- ④ Completely rewritten
 - ④ Rich UI actions, drag and drop
 - ④ Backlog and Task Board
- ④ Real-time charts
 - ④ Velocity and Sprint burndown
 - ④ Set the Start and End dates
- ④ Forecast Tool
 - ④ Use velocity for release planning
- ④ Three License modes
 - ④ Limited, Standard, Full



Teams

- ④ A Team is a new concept in TFS 11
 - ④ Description & photo
- ④ Default team
- ④ Security
- ④ Scoping Teams
- ④ Team Alerts
- ④ Team Favorites

The screenshot shows the 'Tailspin Team' overview page in TFS 11. The breadcrumb navigation at the top reads: Control Panel > Samples > Tailspin > Tailspin Team. Below this is a tab bar with 'overview', 'security', and 'alerts'. The 'overview' tab is active, showing a 'team profile' section on the left and a 'team members' section on the right.

team profile



Name
Tailspin Team

Description
The default project team.

Administrators
Administrator *

[+ add](#)

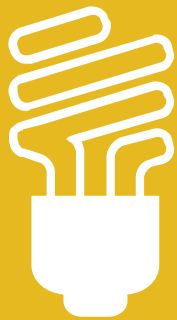
team members

[Add...](#) | 

Display Name	Username or Scope
 Administrator	VSTFS\Administrator
 Dave (Developer)	VSTFS\Dave
 Developers	[VSTFS]
 Dieter (Developer)	VSTFS\Dieter
 Jake (Stakeholder)	VSTFS\Jake
 Scott (Scrum Master)	VSTFS\Scott
 Toni (Developer)	VSTFS\Toni
 Wade (Developer)	VSTFS\Wade

Activities: Setting up a new ~~Team Project~~ Product

- ④ Create the team project
 - ④ Use the Visual Studio Scrum 2.0 process template
- ④ Configure security groups, permissions, team, photos
 - ④ I'm a fan of the Scrum Team being project administrators
- ④ Configure the work item areas and iterations
 - ④ If there are multiple teams, scope ownership by area
- ④ Publish the Definition of "Done"
 - ④ SharePoint works well here
- ④ Migrate any work items



Setting up a new Scrum project

Mapping Scrum to Visual Studio 2012

*

Scrum Concept	Visual Studio Concept
Team	Team Foundation Server security groups
Definition of Done	Wiki entry or document on the portal
Product	Team Project
Product Backlog	Product Backlog shared query or Backlog page on TWA
Product Backlog Item	PBI or Bug work item
Acceptance Criteria	Acceptance Criteria field and (optionally) associated Test Case work item(s)
Release	Iteration Path (high level)
Release Plan	Backlog page on Team Web Access (w/ Forecast tool)
Sprint	Iteration Path (low level)
Sprint Dates	Iteration start and end dates
Sprint Goal	Wiki entry or document on the portal
Sprint Backlog	Sprint Backlog shared query or Backlog & Board pages on TWA
Task	Task work item
Burndown and Velocity	Release Burndown, Sprint Burndown, and Velocity reports and charts on TWA
Impediment	Impediment work item
Retrospective	Wiki entry or document on the portal



The Product Backlog



(It's your list of desirements)

Example Product Backlog

Anyone can provide this

Product Owner provides this

Product Owner provides this

Feature

Bug

Epic

ID	Title	Description	Area	Value	Effort	Priority
1	Customer Login	As a returning ...	Admin	45	8	2
2	Product return	As an unhappy ...	Products	75	13	4
3	Twitter feed	As a visitor to ...	Marketing	25	5	3
4	Wrong sales tax	As a purchaser ...	Orders	90	3	1
5	Mobile-friendly site	As a mobile user ...	*	60	34	

Development Team provides this estimate

The Backlog page in Team Web Access

- ② Two views
 - ② Product Backlog
 - ② Sprint Backlog
- ② Hierarchical
- ② Quick Add
- ② Drag and drop
- ② Customizable

The screenshot displays the 'Product Backlog' page in Team Web Access. The top navigation bar includes 'HOME', 'WORK', 'SOURCE', and 'BUILD'. Below this, the 'backlog' tab is selected, with 'board' and 'work items' tabs also visible. The left sidebar shows a tree view with 'Product Backlog' expanded, listing 'Current' (Sprint 1) and 'Future' (Sprints 2 through 6). The main content area is titled 'Product Backlog' and contains a 'contents' tab. A 'Quick Add' form is positioned above a table of tasks. The form has a 'Type' dropdown set to 'Product Backlog Item' and a 'Title' input field with an 'Add' button. The table below has columns for 'Order', 'Title', 'State', 'Effort', and 'Iteration Path'. It lists 13 tasks, all with a state of 'Approved' and an iteration path of 'Tailspin'.

Order	Title	State	Effort	Iteration Path
1	Improve UX for mobile devices	Approved		Tailspin
2	Improve home page UX for mobile devices	Approved	8	Tailspin
3	Improve product browsing UX for mobile devices	Approved	5	Tailspin
4	Improve ordering UX for mobile devices	Approved	8	Tailspin
5	Bug: Duplicate product in shopping cart	Approved	5	Tailspin
6	Display product version and web server information	Approved	5	Tailspin
7	Customer login	Approved	13	Tailspin
8	Check order status	Approved	5	Tailspin
9	Return a product	Approved	13	Tailspin
10	Bug: Shipping and handling fees are wrong	Approved	8	Tailspin
11	Bug: Oregon sales tax not computed correctly	Approved	5	Tailspin
12	Exchange a product	Approved	8	Tailspin
13	Feed job openings from http://hrweb portal	Approved	8	Tailspin

The PBI Work Item Type

Product Backlog Item 1 : Add a Twitter feed to the homepage

Add a Twitter feed to the homepage

Iteration Tailspin

STATUS

Assigned To Paula (Product Owner)

State Approved

Reason Approved by the Product Owner

DETAILS

Effort 8

Business Value 25

Area Tailspin\Community

DESCRIPTIONSTORYBOARDSTEST CASESTASKS

ACCEPTANCE CRITERIAHISTORYLINKSATTACHMENTS

Arial 3 **B I U**    

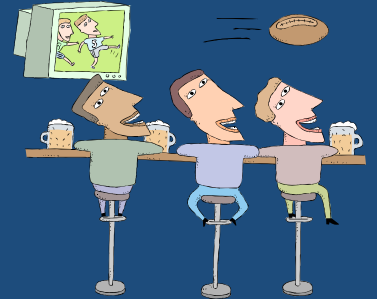
As a visitor to the Tailspin Toys homepage I want to see a list of recent tweets about the company and its products so that I know that they are alive and well.

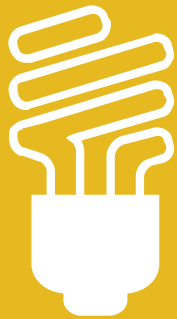
  **B I U**    

- Feed should look and feel like the rest of the page
- Only show the last 10 tweets
- Use the company's Twitter account
- Include tweets with #TailspinToys hashtag
- New tweets should appear <15 minutes after posting
- Filter out spam and profanity

Demo: Creating the Product Backlog

- ④ Use the results of our Product Backlog creation activity
- ④ Create the initial items in the Product Backlog

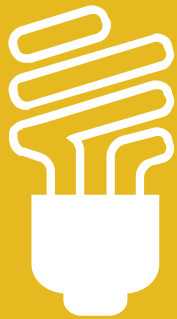




Setting up the Product Backlog

Product Backlog Grooming

- ④ Grooming is the act of adding detail, estimates, and order to items in the Product Backlog
 - This is an ongoing process in which the Product Owner and the Development Team collaborate on the details of Product Backlog items
- ④ The Scrum Guide says grooming can (should) take up to 10% of the length of Sprint
 - Budget the time, schedule the meeting, and attend



Grooming the Product Backlog

Tip: Find a Good Product Owner

- ④ The Product Owner is responsible for maximizing the value of the product and the work of the Development Team
 - ④ In order to do this, he or she needs an ordered list of everything that might be needed in the product
- ④ The Product Owner role has the most responsibility
 - ④ This is the hardest role to fill and is the most demanding





The Sprint



(Achieving our Sprint Goal)

Activities: Setting up a New Sprint

- ④ Ensure that the release and Sprint nodes exist with dates
- ④ Modify the Current Sprint queries (to the current Sprint)
 - ④ Or use fellow PSD Mark Michaelis' cool trick: <http://bit.ly/J7eHHY>
- ④ Run the Product Backlog query for accuracy
- ④ Review the Release Burndown and Velocity reports
- ④ As a result of Sprint Planning:
 - ④ Change Iteration Path to the Sprint
 - ④ Change state to *Committed*
 - ④ Create and link initial task work items
- ④ Review the Sprint Backlog query for accuracy

The (Sprint) Backlog page in TWA

- ⌚ Shows forecasted PBIs & Bugs and linked tasks
- ⌚ What's missing: quick add, drag & drop ordering, bug icons
- ⌚ What's not important: Capacity, work by Activity & Assigned To

Sprint 1						
contents		capacity				
ID	Title	State	Remaining Work	Work Item Type	Assigned To	Effort
+ 5	Add a Twitter feed to the home page	Committed	40	Product Backlog Item	Richard Hundhausen	8
10	Create a Twitter account	To Do	2	Task		
11	Build a control to display the tweets	To Do	8	Task		
12	Create test cases and tests	To Do	8	Task		
13	Create supporting database objects	In Progress	2	Task	Dieter	
14	Code the feed reader	In Progress	16	Task	Jake	
15	Creating a rollout plan and automation to support it	To Do	4	Task		
+ 4	Bug: Items incorrectly added to shopping cart	Committed		Bug	Richard Hundhausen	5
+ 7	Check order status	Committed		Product Backlog Item	Richard Hundhausen	8

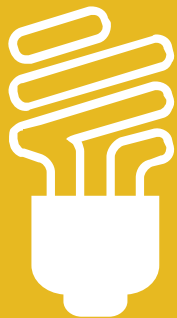
The Task Work Item Type

- Tasks should be small enough to be completed in one day
 - Remaining work is in hours
- It's ok to have unassigned tasks
 - Don't assign tasks
- Avoid using the Activity field
 - All activities are "development"

The screenshot shows a web-based form for creating a task. The title bar reads 'Task 1812: Create SQL Server objects'. The form has a title 'Create SQL Server objects' and an 'Iteration' dropdown set to 'Tailspin\Release 1\Sprint 1'. It is divided into several sections: 'STATUS' with fields for 'Assigned To', 'State' (set to 'To Do'), 'Reason' (set to 'New task'), and 'Blocked'; 'DETAILS' with fields for 'Remaining Work' (24), 'Backlog Priority', 'Activity', and 'Area' (set to 'Tailspin\Orders'); 'DESCRIPTION' with the text 'Create the database, tables, stored procedures, and DAL code.'; and a right-hand pane with tabs for 'HISTORY', 'LINKS', and 'ATTACHMENTS', currently showing 'LINKS' with a table header 'ID | Work Item... | Title'. At the bottom right are buttons for 'Save', 'Save and Close', and 'Cancel'.

STATUS		DETAILS	
Assigned To		Remaining Work	24
State	To Do	Backlog Priority	
Reason	New task	Activity	
Blocked		Area	Tailspin\Orders

DESCRIPTION		LINKS	
Create the database, tables, stored procedures, and DAL code.		ID	Work Item... Title



Creating the Sprint Backlog

The (Task) Board page in TWA

- Visual and manage tasks in the current Sprint
- Drag and drop, (limited) inline editing

The screenshot displays the TWA Board page for Sprint 1, which runs from May 30 to June 12, with 9 work days remaining. The interface includes a top navigation bar with tabs for 'HOME', 'WORK' (selected), 'SOURCE', and 'BUILD'. Below this, there are sub-tabs for 'backlog', 'board' (selected), and 'work items'. A search bar labeled 'Search work items...' is located on the right. The main area shows the 'Sprint 1' board with columns for 'TO DO 22 h', 'IN PROGRESS 18 h', and 'DONE'. A 'backlog items' tab is also visible. The 'TO DO' column contains four tasks: 'Add a Twitter feed to the home page' (40 h), 'Create a Twitter account' (2 h), 'Create test cases and tests' (8 h), and 'Build a control to display the tweets' (8 h). The 'IN PROGRESS' column contains two tasks: 'Create supporting database objects' (2 h, assigned to Dieter) and 'Code the feed reader' (16 h, assigned to Jake). The 'DONE' column is currently empty. A 'person' dropdown menu is set to 'Dieter'. At the bottom, there are two notifications: 'Bug: Items incorrectly added to shopping cart' and 'Check order status'.

TO DO 22 h	IN PROGRESS 18 h	DONE
- Add a Twitter feed to the home page 40 h - Create a Twitter account 2 - Create test cases and tests 8	- Create supporting database objects 2 Dieter - Code the feed reader 16 Jake	

Notifications:

- ⚠ Bug: Items incorrectly added to shopping cart
- ⚠ Check order status

Developer Activities During the Sprint

- ④ Attend Daily Scrum meetings
- ④ Create and link task work items as needed
 - ④ Developers take on (sign up for) tasks from the Sprint Backlog
 - ④ Work is never assigned, even by other developers
- ④ Check-in code changes to TFS associating with one task only
- ④ Update remaining work estimates on your tasks daily
- ④ Complete work according to acceptance criteria and the DoD
 - ④ As you complete a task, change its state and clear remaining work fields
- ④ Groom the Product Backlog by creating and editing PBIs and Bugs
- ④ Create and manage impediment work items as needed
- ④ Review the Sprint Burndown chart to monitor progress

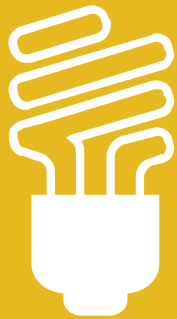
The Bug Work Item Type

- ⌵ A bug work item is the *report* of a bug in the product
- ⌵ The bug work item type is similar to a PBI
- ⌵ In general, bugs should be fixed not managed
- ⌵ **Tip:** Prefix the bug with "Bug:" they are identifiable in the Product Backlog

The screenshot shows a bug work item form titled "Bug 1816: Bug: Oregon sales tax being computed incorrectly". The form includes a title field with the same text, an "Iteration" dropdown set to "Tailspin", and a "STATUS" section with fields for "Assigned To" (Paula (Product Owner)), "State" (New), and "Reason" (New defect reported). A "DETAILS" section contains "Effort", "Severity" (3 - Medium), and "Area" (Tailspin\Orders). Below these are tabs for "STEPS TO REPRODUCE", "SYSTEM", "TEST CASES", "TASKS", "ACCEPTANCE CRITERIA", "HISTORY", "LINKS", and "ATTACHMENTS". The "STEPS TO REPRODUCE" tab is active, showing a list of steps: "1. Add a product to an empty cart", "2. Check out", "3. Specify an Oregon state address", and "4. Observe sales tax". The "ACCEPTANCE CRITERIA" tab shows the text: "* No sales tax should be computed for Oregon residents". At the bottom right are "Save", "Save and Close", and "Cancel" buttons.

Completing a Sprint

- ④ Deliver and demo a “potentially releasable” increment
 - ④ Groom the Product Backlog as needed
- ④ Set the state of all current Sprint work items appropriately
 - ④ Done vs. unfinished/rejected work
- ④ Label code and other artifacts in version control
 - ④ Branch as needed (i.e. to support hotfixing)
- ④ Perform a Sprint Retrospective
 - ④ Capture the feedback appropriately
- ④ Prepare for the next Sprint



Completing a Sprint



Getting to Done



(Don't be flaccid)

Flaccid Scrum

- ④ Martin Fowler describes *flaccid* Scrum <http://bit.ly/17aEU>
- ④ Flaccid teams believe in magic
- ④ Flaccid customers believe in magic
- ④ Don't just inspect ... adapt
- ④ Be professional!

"Done"

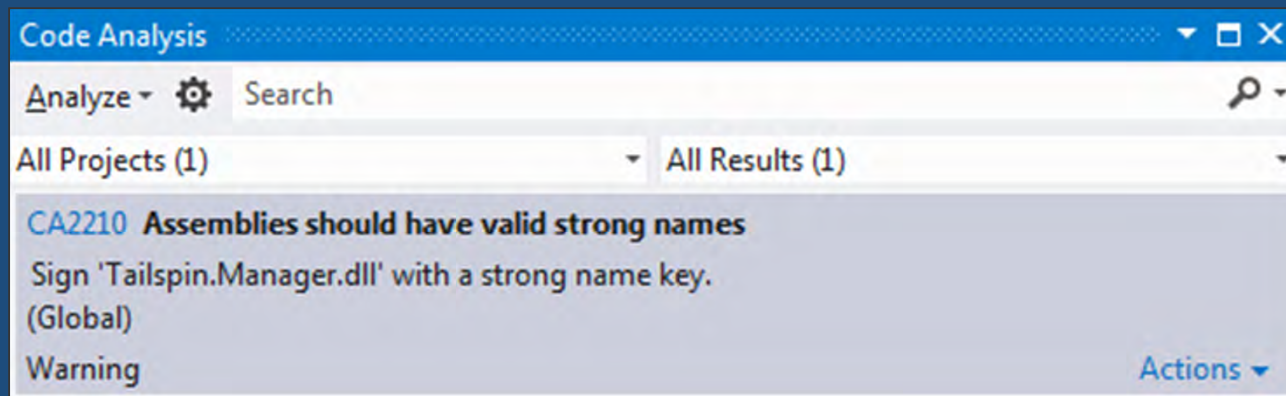
- ④ Done defines when an increment of product functionality is potentially shippable
- ④ Definition of Done (DoD)
 - ④ A simple, auditable checklist owned by the team
 - ④ It can be influenced by organizational standards and specific requirements of the product or release
- ④ Can be maintained ...
 - ④ In a document or wiki on the project portal

A Simple Definition of Done

1. Code complete
2. Tests pass
3. Everything checked-in
4. Acceptance criteria from Product Owner met

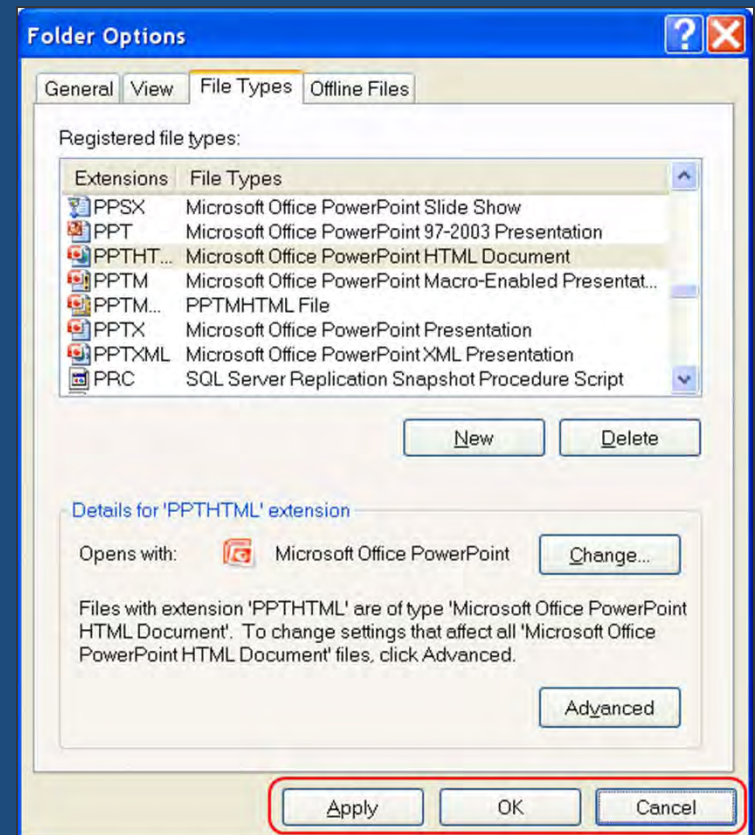
Quiz: What if you are not really done?

Your team believes that they have met their Sprint Goal and the Product Owner is quite happy with the increment's functionality. One of the PBIs wasn't completed according to your definition of done. It seems that a Code Analysis warning remains:



Quiz: What about that dialog window?

Your Product Owner has accepted this dialog box as being fit for purpose, but you are concerned that the buttons at the bottom do not match the rest of those in your application or in Windows. The feature is done and has been accepted.



Done



Learn more here: www.scrum.org

Check out our training services: www.accentient.com

Contact me: richard@accentient.com

Related Content

- ④ DEV212 (Implementing Scrum Using Team Foundation Server 11)
- ④ Hands-on Labs
- ④ Product Demo Stations: Visual Studio ALM / Team Foundation Server
- ④ Related assessment exam: Scrum Open (free) www.scrum.org/scrumopen
- ④ Find me later at: the Visual Studio demo stations and MVP lounge area

Track Resources

⬇ Resource 1

⬇ Resource 2

⬇ Resource 3

⬇ Resource 4

Required Slide

*delete this box when your slide is finalized

Track PMs will supply the content for this slide, which will be inserted during the final scrub.

Resources

*my*tech·ed
Learn. Connect. Share.

Connect. Share. Discuss.

<http://northamerica.msteched.com>

Microsoft | Learning

Microsoft Certification & Training Resources

www.microsoft.com/learning

Microsoft | TechNet

Resources for IT Professionals

<http://microsoft.com/technet>

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Resources for Developers

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Complete an evaluation on CommNet and enter to win!



MS Tag

Scan the Tag
to evaluate this
session now on
myTechEd Mobile



Required Slide

*delete this box when your slide is finalized

Your MS Tag will be inserted here during the final scrub.





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